```
import numpy as np
import pandas as pd
list=[[1,'Smith',50000],[2,'Jones',60000]]
df=pd.DataFrame(list)
df
```

```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 2,\n  \"fields\": [\n
{\n      \"column\": 0,\n      \"properties\": {\n        \"dtype\":
\"number\",\n        \"std\": 0,\n        \"min\": 1,\n
\"max\": 2,\n        \"num_unique_values\": 2,\n        \"samples\":
[\n          2,\n          1\n        ],\n        \"semantic_type\":
\"\",\n        \"description\": \"\"\n      }\n    },\n    {\n
\"column\": 1,\n      \"properties\": {\n        \"dtype\":
\"string\",\n        \"num_unique_values\": 2,\n        \"samples\":
[\n          \"Jones\",\n          \"Smith\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": 2,\n      \"properties\": {\n
\"dtype\": \"number\",\n        \"std\": 7071,\n        \"min\":
50000,\n        \"max\": 60000,\n        \"num_unique_values\": 2,\n
\"samples\": [\n          60000,\n          50000\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    }\n  ]\n}","type":"dataframe","variable_name":"df"}
```

```
df.columns=['Empd','Name','Salary']
df
```

```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 2,\n  \"fields\": [\n
{\n      \"column\": \"Empd\",\n      \"properties\": {\n
\"dtype\": \"number\",\n        \"std\": 0,\n        \"min\": 1,\n
\"max\": 2,\n        \"num_unique_values\": 2,\n        \"samples\":
[\n          2,\n          1\n        ],\n        \"semantic_type\":
\"\",\n        \"description\": \"\"\n      }\n    },\n    {\n
\"column\": \"Name\",\n      \"properties\": {\n        \"dtype\":
\"string\",\n        \"num_unique_values\": 2,\n        \"samples\":
[\n          \"Jones\",\n          \"Smith\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Salary\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\": 7071,\n
\"min\": 50000,\n        \"max\": 60000,\n
\"num_unique_values\": 2,\n        \"samples\": [\n          60000,\n
50000\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"df"}
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
```

```
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
```

```
df=pd.read_csv("/content/50_Startups.csv")
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     object
 4   Profit           50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
```

```
df.head()
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 50,\n  \"fields\": [\n
{\n      \"column\": \"R&D Spend\",\n      \"properties\": {\n
\"dtype\": \"number\",\n        \"std\": 45902.25648230753,\n
\"min\": 0.0,\n        \"max\": 165349.2,\n
\"num_unique_values\": 49,\n        \"samples\": [\n
91992.39,\n          1000.23,\n          0.0\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Administration\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
28017.802755488683,\n        \"min\": 51283.14,\n        \"max\":
182645.56,\n        \"num_unique_values\": 50,\n        \"samples\":
[\n          135495.07,\n          82982.09,\n          115641.28\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    },\n    {\n      \"column\": \"Marketing Spend\",\n
\"properties\": {\n        \"dtype\": \"number\",\n        \"std\":
122290.31072584528,\n        \"min\": 0.0,\n        \"max\":
471784.1,\n        \"num_unique_values\": 48,\n        \"samples\": [\
n          353183.81,\n          172795.67,\n          134050.07\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    },\n    {\n      \"column\": \"State\",\n      \"properties\":
{\n        \"dtype\": \"category\",\n        \"num_unique_values\":
3,\n        \"samples\": [\n          \"New York\",\n
\"California\",\n          \"Florida\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"Profit\",\n      \"properties\":
```

{\n   \"dtype\": \"number\",\n   \"std\":
40306.18033765055,\n   \"min\": 14681.4,\n   \"max\":
192261.83,\n   \"num_unique_values\": 50,\n   \"samples\":
[\n   134307.35,\n   81005.76,\n   99937.59\n
],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n
}\n   }\n]\n}","type":"dataframe","variable_name":"df"}

```
df.tail()
```

{"summary":"{\n \"name\": \"df\",\n \"rows\": 5,\n \"fields\": [\n
{\n   \"column\": \"R&D Spend\",\n   \"properties\": {\n
\"dtype\": \"number\",\n   \"std\": 589.7833681700425,\n
\"min\": 0.0,\n   \"max\": 1315.46,\n
\"num_unique_values\": 4,\n   \"samples\": [\n   1315.46,\
n   542.05,\n   1000.23\n   ],\n
\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\
n   },\n   {\n   \"column\": \"Administration\",\n
\"properties\": {\n   \"dtype\": \"number\",\n   \"std\":
32849.625535718515,\n   \"min\": 51743.15,\n   \"max\":
135426.92,\n   \"num_unique_values\": 5,\n   \"samples\": [\
n   115816.21,\n   116983.8,\n   135426.92\n
],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n
}\n   },\n   {\n   \"column\": \"Marketing Spend\",\n
\"properties\": {\n   \"dtype\": \"number\",\n   \"std\":
129061.69240878528,\n   \"min\": 0.0,\n   \"max\":
297114.46,\n   \"num_unique_values\": 4,\n   \"samples\": [\
n   297114.46,\n   45173.06,\n   1903.93\n
],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n
}\n   },\n   {\n   \"column\": \"State\",\n   \"properties\":
{\n   \"dtype\": \"string\",\n   \"num_unique_values\": 3,\n
\"samples\": [\n   \"New York\",\n   \"Florida\",\n
\"California\"\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   },\n   {\n   \"column\":
\"Profit\",\n   \"properties\": {\n   \"dtype\": \"number\",\n
\"std\": 18486.05628017047,\n   \"min\": 14681.4,\n
\"max\": 64926.08,\n   \"num_unique_values\": 5,\n
\"samples\": [\n   49490.75,\n   14681.4,\n
42559.73\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   }\n]\n}","type":"dataframe"}

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/employee.csv")
df.head()
```

{"summary":"{\n \"name\": \"df\",\n \"rows\": 400,\n \"fields\": [\
n   {\n   \"column\": \"emp id\",\n   \"properties\": {\n
\"dtype\": \"number\",\n   \"std\": 115,\n   \"min\": 1,\n
\"max\": 400,\n   \"num_unique_values\": 400,\n
\"samples\": [\n   210,\n   281,\n   34\

```
n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"name\",\n      \"properties\": {\n        \"dtype\": \"string\",\n
\"num_unique_values\": 300,\n        \"samples\": [\n          \"KAVYA
M\",\n          \"KAVYA S\",\n          \"SRI B\"\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"salary\",\n      \"properties\":
{\n        \"dtype\": \"number\",\n        \"std\": 1811,\n
\"min\": 3000,\n        \"max\": 10000,\n
\"num_unique_values\": 8,\n        \"samples\": [\n          6000,\n
4000,\n          5000\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\
n}","type":"dataframe","variable_name":"df"}
```


```
df.tail()
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 5,\n  \"fields\": [\n
{\n      \"column\": \"emp id\",\n      \"properties\": {\n
\"dtype\": \"number\",\n        \"std\": 1,\n        \"min\": 396,\n
\"max\": 400,\n        \"num_unique_values\": 5,\n        \"samples\":
[\n          397,\n          400,\n          398\n        ],\n
\"semantic_type\": \"\",\n        \"description\": \"\"\n      }\
n    },\n    {\n      \"column\": \"name\",\n      \"properties\": {\n
\"dtype\": \"string\",\n        \"num_unique_values\": 5,\n
\"samples\": [\n          \"TANVI V\",\n          \"YASH C\",\n
\"UMESH A\"\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    },\n    {\n      \"column\":
\"salary\",\n      \"properties\": {\n        \"dtype\": \"number\",\n
\"std\": 2588,\n        \"min\": 3000,\n        \"max\": 10000,\n
\"num_unique_values\": 5,\n        \"samples\": [\n          10000,\n
7000,\n          5000\n        ],\n        \"semantic_type\": \"\",\n
\"description\": \"\"\n      }\n    }\n  ]\n}","type":"dataframe"}
```


```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 400 entries, 0 to 399
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  400 non-null    int64
 1   name    400 non-null    object
 2   salary  400 non-null    int64
dtypes: int64(2), object(1)
memory usage: 9.5+ KB
```

```
df.salary
```

```
0       5000
1       6000
2       7000
```

```
3       5000
4       8000
       ...
395     3000
396    10000
397     5000
398     6000
399     7000
Name: salary, Length: 400, dtype: int64
```

```
type(df.salary)
```

```
pandas.core.series.Series
```

```
df.salary.mean()
```

```
np.float64(6005.0)
```

```
df.salary.median()
```

```
6000.0
```

```
df.salary.mode()
```

```
0    5000
1    6000
Name: salary, dtype: int64
```

```
df.salary.var()
```

```
3283182.9573934837
```

```
df.salary.std()
```

```
1811.9555616497562
```

```
df.describe()
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 8,\n  \"fields\": [\n{\n      \"column\": \"emp id\",\n      \"properties\": {\n\"dtype\": \"number\",\n        \"std\": 143.85727156983504,\n\"min\": 1.0,\n        \"max\": 400.0,\n        \"num_unique_values\": 6,\n        \"samples\": [\n          400.0,\n          200.5,\n300.25\n        ],\n        \"semantic_type\": \"\",\n\"description\": \"\"\n      }\n    },\n    {\n      \"column\": \"salary\",\n      \"properties\": {\n        \"dtype\": \"number\",\n\"std\": 3076.5706213445333,\n        \"min\": 400.0,\n\"max\": 10000.0,\n        \"num_unique_values\": 8,\n\"samples\": [\n          6005.0,\n          6000.0,\n          400.0\n        ],\n        \"semantic_type\": \"\",\n\"description\": \"\"\n      }\n    }\n  ]\n}","type":"dataframe"}
```


```
df.describe(include='all')
```

{"summary":"{\n \"name\": \"df\",\n \"rows\": 11,\n \"fields\": [\n{\n   \"column\": \"emp id\",\n   \"properties\": {\n\"dtype\": \"number\",\n   \"std\": 143.85727156983504,\n\"min\": 1.0,\n   \"max\": 400.0,\n   \"num_unique_values\": 6,\n   \"samples\": [\n   400.0,\n   200.5,\n300.25\n   ],\n   \"semantic_type\": \"\",\n\"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"name\",\n   \"properties\": {\n   \"dtype\": \"category\",\n\"num_unique_values\": 4,\n   \"samples\": [\n   300,\n\"3\",\n   \"400\"\n   ],\n   \"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n\"column\": \"salary\",\n   \"properties\": {\n   \"dtype\": \"number\",\n   \"std\": 3076.5706213445333,\n   \"min\": 400.0,\n   \"max\": 10000.0,\n   \"num_unique_values\": 8,\n\"samples\": [\n   6005.0,\n   6000.0,\n   400.0\n   ],\n   \"semantic_type\": \"\",\n\"description\": \"\"\n   }\n   }\n]\n}","type":"dataframe"}

```
empCol=df.columns
empCol
```

```
Index(['emp id', 'name', 'salary'], dtype='object')
```

```
emparray=df.values
emparray
```

```
array([[1, 'ANANYA R', 5000],
       [2, 'BHAVESH K', 6000],
       [3, 'SREYA G', 7000],
       ...,
       [398, 'UMESH A', 5000],
       [399, 'VARUN B', 6000],
       [400, 'YASH C', 7000]], dtype=object)
```

```
employee_DF=pd.DataFrame(emparray,columns=empCol)
employee_DF
```

{"summary":"{\n \"name\": \"employee_DF\",\n \"rows\": 400,\n\"fields\": [\n   {\n   \"column\": \"emp id\",\n\"properties\": {\n   \"dtype\": \"date\",\n   \"min\": 1,\n\"max\": 400,\n   \"num_unique_values\": 400,\n\"samples\": [\n   210,\n   281,\n   34\n   ],\n   \"semantic_type\": \"\",\n\"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"name\",\n   \"properties\": {\n   \"dtype\": \"string\",\n\"num_unique_values\": 300,\n   \"samples\": [\n   \"KAVYA M\",\n   \"KAVYA S\",\n   \"SRI B\"\n   ],\n\"semantic_type\": \"\",\n   \"description\": \"\"\n   }\n   },\n   {\n   \"column\": \"salary\",\n   \"properties\": {\n   \"dtype\": \"date\",\n   \"min\": 3000,\n\"max\": 10000,\n   \"num_unique_values\": 8,\n

```
\"samples\": [\n          6000,\n          4000,\n          5000\n
],\n        \"semantic_type\": \"\",\n        \"description\": \"\"\n
}\n    }\n  ]\n}","type":"dataframe","variable_name":"employee_DF"}
```